PUSL3190 Computing Project

Project Initiation Document (PID)

AI-based career guidance system

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Chapter 01: Introduction

1.1: Background and Context

Sri Lanka has consistently achieved impressive educational outcomes with literacy rates exceeding 92% and school enrollment rates of 97%. However, youth unemployment remains persistently high at 16.7%, approximately four times the national average revealing a critical gap between educational achievement and workforce integration.(Department of Census and Statistics, 2023; World Bank, 2022).
Traditional career counseling services in Sri Lanka are severely limited, particularly in rural areas(ILO, 2015). The International Labour Organization explicitly recognizes that inadequate career counseling contributes significantly to high youth unemployment rates. Most Sri Lankan youth receive career guidance primarily from parents and teachers rather than professional career services, reflecting insufficient support infrastructure.

A fundamental problem is the skills mismatch between educational outputs and labor market demands. Students often select educational pathways based on institutional prestige or family influence rather than personal aptitude or market realities, resulting in poor career outcomes. The G.C.E. Advanced Level examination channels students into narrow career perspectives dominated by medicine, engineering, and government employment, overlooking emerging opportunities in technology, entrepreneurship, and vocational sectors(World Bank, 2022).

1.2 Problem Statement

School leavers in Sri Lanka face a complex decision-making environment when transitioning from education to career paths. The primary challenges include:

- **Information Asymmetry:** Students lack comprehensive knowledge about diverse career options beyond conventional fields, leading to uninformed decisions misaligned with their abilities and interests.
- **Absence of Personalized Guidance:** Current systems provide generic advice that fails to consider individual academic performance, skills, personality traits, and aspirations.
- **Disconnect from Market Realities:** Educational choices are made without understanding current job market demands, causing graduates to pursue oversaturated fields while emerging sectors face talent shortages.
- **Geographic Disparities:** Rural and economically disadvantaged students have severely limited access to quality career guidance services.

These challenges result in prolonged unemployment, underemployment, career dissatisfaction, and brain drain. The economic cost includes wasted educational investments (LKR 500,000-1,000,000 per unsuitable degree), foregone earnings, and hindered national development.

1.3 Scope and Limitations

In Scope:

- Accept user input including academic records (G.C.E. O/L and A/L results), educational stream, certifications, skills, and personal interests.
- Generate personalized career and higher education recommendations using machine learning algorithms.
- Integrate real-time job market data from Sri Lankan sources including the Department of Census & Statistics, ICTA databases, and online job portals.
- Provide information about universities, vocational institutes, and short courses aligned with recommended career paths.
- Multilingual support (Sinhala, Tamil, English)
- Career comparison and exploration features.

Out of Scope:

- Direct job placement or recruitment services.
- Comprehensive psychometric testing requiring clinical validation.
- Mobile native applications (limited to responsive web design)

Limitations:

- Recommendation accuracy depends on data quality and availability
- Cannot fully replace human counseling for complex circumstances
- Initial focus on major career domains, not highly specialized niches

1.4 Expected Impact and Stakeholders

Primary Stakeholders:

- School Leavers (Ages 16-20): Direct beneficiaries who will use the system to make informed educational and career decisions
- Parents and Guardians: Indirect users seeking to support their children's career planning
- School Career Guidance Counselors: Potential adopters who can use the platform as a supplementary tool

Secondary Stakeholders:

- Educational Institutions: Universities and vocational training centers seeking to better understand student aspirations

- Ministry of Education, Sri Lanka: Policy-making body interested in evidence-based career guidance solutions
- ICTA: Technology partner providing data access and validation
- Private Sector Employers: Organizations benefiting from better-prepared candidates

Expected Impacts:

- Individual Level: Increased awareness of diverse career options, improved decision-making confidence, reduced time-to-employment
- Institutional Level: Enhanced efficiency of career guidance services, scalable solution reaching underserved communities
- Societal Level: Reduced youth unemployment and underemployment, better alignment between educational outputs and economic needs

Chapter 02: Business Case

2.1 Business Need

According to the Department of Census and Statistics and World Bank reports, Sri Lanka faces a critical human capital challenge where qualified youth remain unemployed while industries report skill shortages. Research shows most youth receive inadequate career guidance, with unsuitable general education, lack of English proficiency, and absence of influential networks perceived as major employability obstacles.

Manual career counseling costs LKR 2,000-5,000 per session with limited availability. An AI-based system provides 24/7 access at marginal cost per user, potentially serving thousands simultaneously and eliminating geographic barriers. If the system helps even 10% of users avoid career mismatches, the aggregated savings (preventing wastage of LKR 500,000-1,000,000 per unsuitable degree) justify development investment.

2.2 Business Objectives

Objective 1: Enhance Accessibility

- Target: Make career guidance accessible to at least 10,000 school leavers within the first year of deployment
- Measurement: Number of registered users, geographic distribution analysis, usage frequency metrics

Objective 2: Improve Decision Quality

- Target: Achieve 80% user satisfaction rating regarding relevance and usefulness of recommendations
- Measurement: Post-usage surveys, feedback ratings, follow-up studies on actual career choices versus recommendations

Objective 3: Reduce Career Mismatch

- Target: 30% decrease in career dissatisfaction versus non-users
- Measurement: Comparative studies tracking career outcomes of system users versus control groups

Objective 4: Provide Market Intelligence

- Target: Integrate ≥5 major Sri Lankan data sources
- Measurement: Number of data sources integrated, recommendation accuracy based on actual job availability

Objective 5: Technical Performance

- Target: System response time under 3 seconds, 99.5% uptime, recommendation generation accuracy ≥85% when validated against expert counselor assessments
- Measurement: System performance monitoring, A/B testing with professional career counselors

Chapter 03: Project Objectives

Objective 1: Develop AI Recommendation Engine

The AI recommendation engine shall provide career suggestions that align with user profiles and demonstrate validity when compared against expert career counselor assessments. The system's recommendations should reflect genuine career matches based on academic performance, skills, and interests rather than generic suggestions.

- The system successfully processes user input data (academic records, skills, interests)
- Recommendations are generated and ranked within 3 seconds
- At least 50 distinct career categories can be recommended

Objective 2: Create User-Friendly Web Application Interface

The web application shall be intuitive and easy to navigate for school leavers with varying levels of technical proficiency. Users should be able to complete core tasks such as profile creation and viewing recommendations without external assistance or extensive training.

- Users can create a complete profile in under 5 minutes
- Interface adapts responsively to screen sizes from 320px (mobile) to 1920px+ (desktop)
- All core features (profile creation, recommendations, exploration, comparison) are fully functional

Objective 3: Integrate Labor Market Data

Labor market information and job availability data shall remain current and relevant. The system should reflect recent trends and changes in the employment landscape to provide users with actionable intelligence.

- Automated ETL pipeline successfully connects to Department of Census & Statistics
- ICTA database integration is operational

Objective 4: Implement Multilingual Support

Multilingual content shall maintain semantic accuracy and cultural appropriateness across Sinhala, Tamil, and English versions. Technical terminology related to careers and education shall be translated correctly to ensure users understand recommendations regardless of their language preference.

- Complete user interface is available in Sinhala, Tamil, and English
- Users can switch languages seamlessly without losing session data
- All career descriptions, educational pathway information, and system messages are translated
- Language preference persists across user sessions

Objective 5: Secure Data Management

The system shall protect user data from unauthorized access, breaches, and vulnerabilities. Security measures should comply with recognized standards and best practices for web application security.

- User authentication system prevents unauthorized access
- User data stored in Firebase is encrypted at rest

Chapter 04: Literature Review

4.1 Technological Approaches

Research demonstrates that AI-powered career guidance systems analyze multifaceted student profiles to provide tailored recommendations. Systems utilizing machine learning and NLP have achieved 85% alignment with expert-validated suggestions, 90% user satisfaction, and 1.5-second response times, establishing performance benchmarks (Kardan & Ebrahimi, 2013; Aggarwal, 2016).

Recommendation Algorithms: Collaborative filtering identifies patterns in user-item interactions but faces cold start problems with new users lacking historical data. Content-based filtering examines career characteristics independent of user behaviors, avoiding cold start issues. Hybrid approaches combining both methods yield superior results by leveraging complementary strengths. Advanced systems use Random Forest, Naive Bayes, and neural networks with NLP techniques for improved accuracy (Bobadilla et al., 2013).

4.2 Sri Lankan Context

The ILO recognizes that lack of career counseling contributes significantly to Sri Lanka's high youth unemployment despite high literacy rates. Studies reveal most youth consider unsuitable general education the main employability obstacle, with school-to-work transitions difficult due to skills mismatches. Career guidance infrastructure is severely under-resourced, particularly in rural areas, with geographic isolation and ICT gaps creating access barriers (International Labour Organization 2015).

Female youth unemployment is approximately double male rates. Students from disadvantaged schools face additional challenges including inadequate finances, poor guidance services, and weak academic performance, necessitating inclusive design approaches addressing diverse needs.

4.3 Design Considerations

Ethical AI deployment requires anonymized data and regular bias audits for gender and socioeconomic factors. Successful systems emphasize transparency users should understand why recommendations were made. Explainability builds trust and enables critical evaluation. Integration of heterogeneous data sources with varying formats and reliability presents significant engineering challenges requiring robust validation(Arrieta, A. B., et al. 2020)

4.4 Research Gaps

Most existing systems target developed countries with different educational structures. Few address Sri Lankan G.C.E. systems or provide comprehensive Sinhala/Tamil support. Many rely on static databases rather than dynamic market data integration. Few explicitly serve rural or economically disadvantaged populations.

This project addresses these gaps through Sri Lankan-calibrated algorithms, trilingual support, dynamic data integration, and equity-focused design contributing to limited South Asian career guidance technology research (IJSRD 2024).

Chapter 05: Method of Approach

5.1 Methodology

The project employs design science research methodology with Agile development using two-week sprints over 30 weeks across six phases: Research & Planning (Weeks 1-4), System Design (5-8), Core Development (9-16), Feature Development (17-22), Testing & Evaluation (23-26), and Finalization (27-30).

5.2 Data Sources

User Data: Academic records (G.C.E. O/L and A/L), skills, interests, career aspirations
Labor Market Data: Department of Census & Statistics (Labour Force Survey), ICTA (tech sector data), job portals (TopJobs.lk, Ikman.lk, LinkedIn), UGC (university programs), Vocational Training Authority (courses)
Collection Plan: 5,000+ career profiles, 10,000+ job listings, 500+ educational programs via web scraping, APIs, and manual curation. Weekly job data updates, quarterly educational data updates. All personal data collected with consent, anonymized, and securely stored.

5.3 AI Model Development

Hybrid Recommendation System:

- Content-Based Filtering (40%): Feature extraction, user vectorization, similarity computation handles new users
- Collaborative Filtering (30%): Matrix factorization using SVD discovers non-obvious matches
- Knowledge-Based Rules (30%): Domain expert logic ensures baseline quality

Technology: Python 3.9+, Scikit-learn, TensorFlow/Keras, Pandas, NumPy, NLTK/spaCy
Training: 70% training, 15% validation, 15% test split with cross-validation and hyperparameter tuning
Evaluation: Precision/Recall/F1-score, Mean Average Precision, NDCG, coverage, diversity, novelty metrics

5.4 System Architecture

Three-Tier Architecture:

- Presentation: React.js 18.x with Tailwind CSS, Redux, Recharts, react-i18next
- Application: Node.js/Express, RESTful API, JWT authentication, Winston logging
- Data: Firebase Realtime Database/Firestore, Redis caching, Elasticsearch search

Deployment: Firebase Hosting (frontend), Google Cloud Run (backend), GitHub Actions (CI/CD), Google Cloud Monitoring.

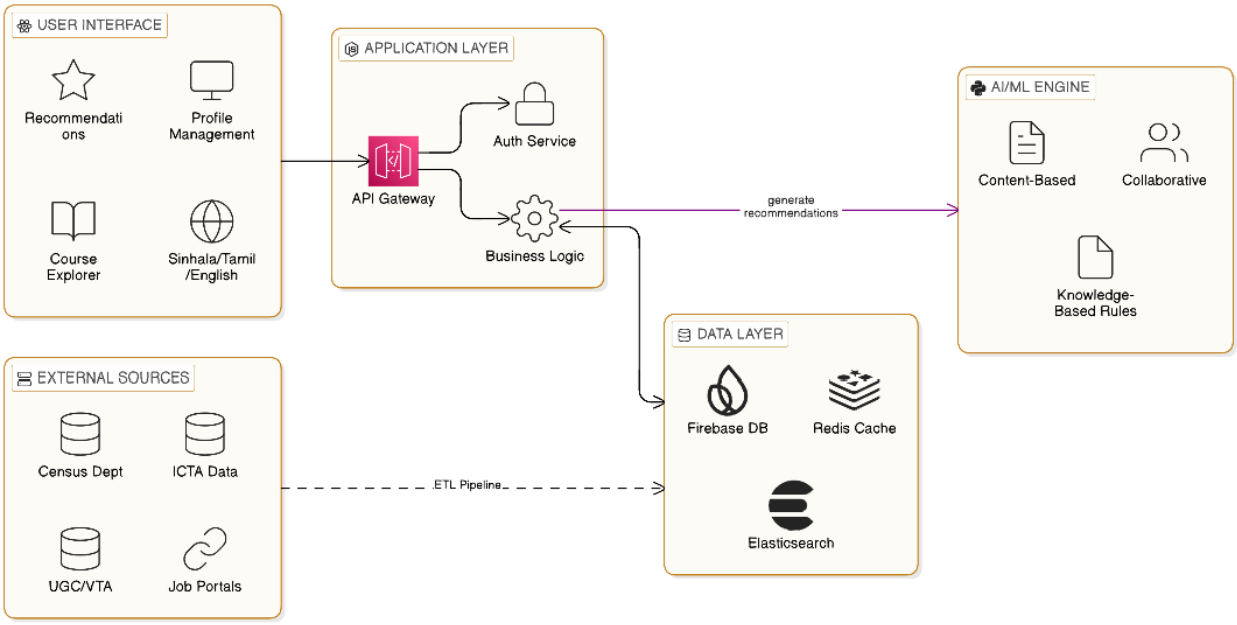
5.5 Evaluation Strategy

Technical: Unit testing (Jest, Mocha/Chai, >80% coverage), integration testing (Supertest), performance testing (JMeter), security testing (OWASP ZAP)

User Evaluation:

- Quantitative: SUS survey (≥75 target), task completion rate (≥90%), time-on-task, acceptance rate
- Qualitative: Semi-structured interviews (20 users), think-aloud protocols, focus groups with counselors
- Validation: Expert review by career counselors, comparative analysis vs. traditional methods, 3-month pilot deployment

Chapter 06: Conceptual Diagram



Chapter 07: Initial Project Plan

Phase	Weeks	Key Activities	Deliverables
Research & Planning	1-4	Literature review, requirements, stakeholder consultation	Requirements Doc, Proposal
System Design	5-8	Architecture, database schema, UI/UX mockups	Architecture Diagram, DB Schema, Mockups
Core Development	9-16	Backend API, authentication, frontend, basic AI model	Functional API, Profile UI, Basic Model
Feature Development	17-22	Assessment integration, data pipeline, multilingual support	Complete Features, Populated DB
Testing & Evaluation	23-26	Unit/integration/performance testing, user validation, model tuning	Test Reports, Improved Model
Finalization	27-30	Bug fixes, deployment, documentation	Deployed App, Final Report

Chapter 08: Risk Analysis

Risk	Likelihood	Impact	Mitigation
Data Quality Issues	High	High	Robust validation, cross-referencing, manual verification
Low User Adoption	Medium	High	User research, student involvement, pilot testing
AI Inaccuracy	Medium	High	Established algorithms, ensemble approach, expert validation
Technical Complexity	Medium	Medium	Documented frameworks, buffer time, mentorship
Scope Creep	Medium	Medium	Clear boundaries, Agile sprints, change control
Privacy Concerns	Low	High	Encryption, OWASP guidelines, informed consent
Integration Challenges	Medium	Medium	Simpler integrations first, error handling, fallbacks
Timeline Delays	Medium	Medium	Buffer time, prioritization, regular monitoring
Limited Market Data	High	Medium	Early partnerships, web scraping, historical baseline
Insufficient Testing	Medium	Medium	Volunteer recruitment, online recruitment, incentives

Contingency Plans: Manual data entry if integration fails; rule-based fallback if AI underperforms; MVP prioritization if timeline exceeded; rapid prototyping cycle for usability issues.

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